

DKA

Diabetic ketoacidosis

DIAGNOSIS AND TEATEMENT

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DKA BY DR. BAKUNDA

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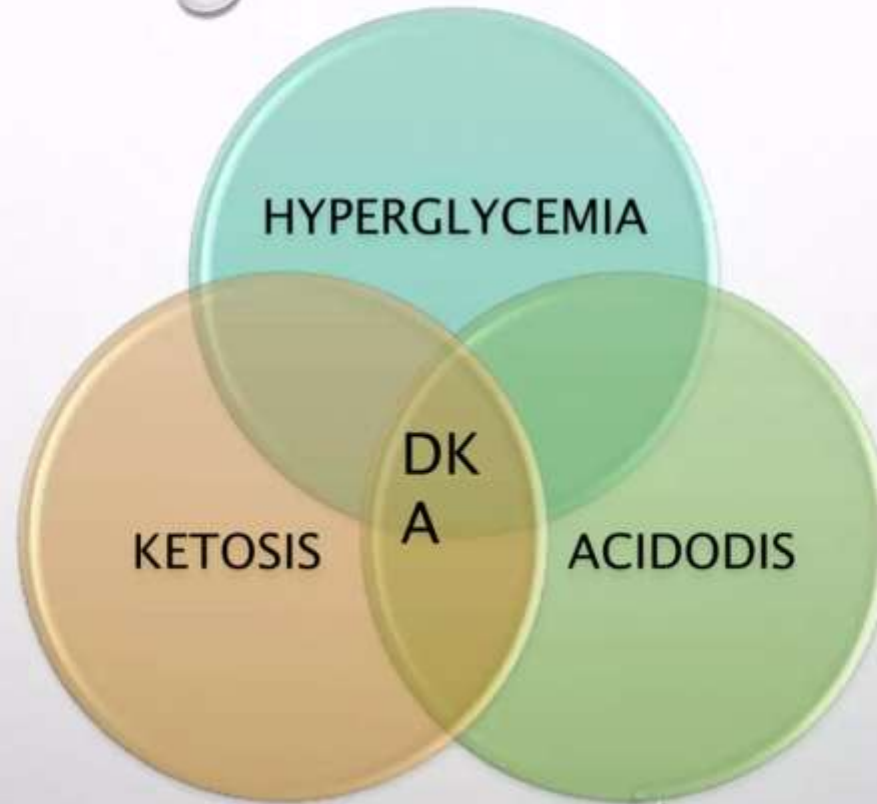


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DEFINITION



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- Diabetic ketoacidosis (DKA) is an acute, major, life-threatening complication of diabetes that mainly occurs in patients with type 1 diabetes, but it can occur in some patients with type 2 diabetes.
 - DKA is a complex disordered metabolic state characterized by hyperglycemia, ketoacidosis, and ketonuria.
 - DKA has high morbidity and mortality in paediatric patients.



Pathophysiology

- Insulin normally elevates cellular uptake of glucose from the blood
- Insulin deficiency with rised counter regulatory hormones (glucagon, cortisol, catecholamines, GH)
- Can occur with lack of insulin (non-adherence, inadequate dosage, 1st presentation) or increased stress (surgery, infection, exercise)



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- Unopposed hepatic glucose production
hyperglycemia cause osmotic diuresis
then dehydration and electrolyte
disturbance Na^+ (water shift to ECF
causing pseudohyponatremia)
 - Fat mobilization increases FFA then
ketoacids then metabolic acidosis



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- As a result of hyperglycemic hyperosmolality, potassium shifts along with water from inside cells to the extracellular space and is lost in the urine.
 - Insulin normally promotes cellular potassium uptake but is absent in DKA, compounding the problem.



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- A total body potassium deficit therefore develops in the body, although serum potassium may be normal or even paradoxically elevated.

- DKA may be precipitated by an intercurrent illness because of failure to increase insulin to compensate for the stress response.
 - Infections (pneumonia, UTI, URI, meningitis, cholecystitis, pancreatitis)
 - Vascular disorder (MI, CVD)
 - Endocrine disorders (hyperthyroidism, cushing's, acromegaly, pheochromocytoma)
 - Trauma
 - Pregnancy
 - Emotional stress
 - Drugs- cocaine



Signs and symptoms

- **Polyuria ,Polydipsia**
- **Recent weight loss**
- **Nausea and vomiting**
- **Signs of volume depletion** (i.e., dry mucous membranes, decreased skin turgor), *hypotension*, circulatory collapse



Neurological abnormalities

- **Altered mental status**
- **Lethargy**
- **coma**
- Other neurological exam abnormalities such as blurred vision and weakness



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- **Rapid onset (< 24 h)**
 - **Abdominal pain**
 - **Fruity odor on the breath** (from exhaled **acetone**)
 - **Hyperventilation: Kussmaul respirations:** deep breaths at a normal respiratory rate



Diagnostic

Initial approach

- ABCs: airway, breathing, and circulation
- Mental status check
- **Fingerstick glucose**
- History (taken from family members or friends if need be), including possible precipitating factors



Diagnostic tests

- ↑ Serum glucose(< 600 mg/dL > 250 mg/dL ,Urine ketones

Serum electrolytes, including bicarbonate (with anion gap calculation)

- Hypertonic hyponatremia
- But if fluid loss due to osmotic diuresis is not replaced, hypernatremia
- serum bicarbonate is reduced
- Urine dipstick with increased glucose and ketones
- Osmolarity (increased to >290 mOsm/L)



Clinical Assessment

Symptoms :

- Polyuria, thirst
- Weight loss
- Weakness
- Nausea, vomiting
- Leg cramps
- Blurred vision
- Abdominal pain

Signs :

- Dehydration
- Hypotension (postural/supine)
- Cold extremities/peripheral cyanosis
- Tachycardia
- Air hunger (Kussmaul breathing)
- Smell of acetone
- Hypothermia
- Hyporeflexia (decreased potassium)
- Confusion, drowsiness, coma (10%)

- **Cardinal biochemical features are:**

- Plasma glucose >250mg/dl
- Arterial pH <7.30
- Serum bicarbonate level <15 mEq/l
- Hyperketonemia >3mmol/L
- Ketonuria >2+ on std urine sticks



Management

Principles

- Admission in high dependency area of Medical Ward or ICU.
- Correction of fluid loss with intravenous fluids
- Correction of hyperglycemia with insulin
- Correction of electrolyte disturbances, particularly hypokalemia
- Correction of acid-base balance but most of time corrected with above mentioned measures
- Treatment of concurrent infection, if present



Correction of fluid loss

- **Normal Saline** or Ringer's Lactate
- Administer 1-3 L during the first hour
- Administer 1 L during the second hour
- Administer 1 L during the following 2 hours
- Administer 1 L every 4 hours, depending on the degree of dehydration



Correction of hyperglycemia

critical to resolve acidosis, not only hyperglycemia

Do not use with hypokalemia , until serum K⁺ is corrected to >3.3 mmol/L

- short-acting insulin(R)
- maintain on 0.1 U/kg/h insulin R infusion
- check serum glucose hourly

If blood glucose stable and urine ketones negative, then stop insulin infusion and start standard insulin regimen



K⁺ replacement

with insulin administration, hypokalemia may develop

- if serum K⁺ < 3.3 mmol/L, hold insulin and give 40 mEq/L K⁺ replacement
- if serum K⁺ 3.3-4.5 mmol/L, give 20 mEq/L K⁺ replacement
- when K⁺ 4.5-6.0 mmol/L add KCL 10 mEq/L IV fluid to keep K⁺ in the range of 3.5-5 mEq/L



Treatment of intercurrent infection

Start empiric antibiotics
on suspicion of infection
until culture results are
available



Complications of DKA

- Metabolic
 - Severe acidosis, hypokalemia, hypoglycemia, hypocalcemia.
- Non metabolic
 - Infection, septic shock, vascular thrombosis, pulm edema, cerebral edema, lactic acidosis, arterial thrombosis, rebound ketoacidosis (premature cessation of insulin therapy).